

SYLLABUS :-

1. Name of the subject Transportation Engineering .New Number. 2. L-T-P 4-0-0
3. Credits 4 4. Name of the Department Department of Civil Engineering 5.
Status of the subject .a. Specify the Session, 2021-22 .Autumn. .b. Please
specify the level of the subject : UG .c. Whether the subject will be offered
as core or elective : Core .d. The semester in which the subject will be
offered: Autumn .e. Name of the programme in whose curricula this subject will
be included : B. Tech 6. Prerequisites for the subject, if any NIL 7.
Objectives Transportation infrastructure plays a key role in the economic
development of a country. It is essential that different components of
transportation infrastructure are planned, designed, built and maintained in an
optimal manner. The students will be introduced to different essential
components of road, road and air transportation and the principles involved in
the estimation of demand for transportation, and the design of different
elements of road, rail and air transportation. 8. Contents 3 Introduction to
Transportation Engineering: Need for Transportation System, Transport Demand
and Supply, Passenger Transportation/Goods Transportation, Different Modes of
Transportation, Urban Transportation/Rural Transportation, 2 Road
Infrastructure Development Process: Prioritization, Feasibility Study,
Preliminary Project Report, Detailed Project Report. 2 Selection of Highway
Alignment: Alternative alignments and evaluation 7 Geometric Design of Roads-I:
Functional Classification, Design Elements and Controls, Cross Section
Elements, Stopping Sight Distance, Overtaking Sight Distance, Decision Sight
Distance, Intermediate Sight Distance, Headlight Sight Distance, Intersection
Sight Distance 6 Geometric Design of Roads-II: Horizontal Alignment- Horizontal
Curve, Superelevation, Transition Curve, Extra Widening, Setback Distance 4
Geometric Design of Roads-III: Topography, Grade Compensation, Critical Length
of Grade, Summit Curve, Valley Curve, Coordination of Horizontal and Vertical
Alignments 5 Traffic Engineering-I: Components of traffic system, Traffic
Stream Characteristics- Uninterrupted Flow Facilities, Interrupted Flow
Facilities, Macroscopic and Microscopic parameters, Fundamental Relationships,
Passenger Car Equivalency/Unit, 5 Traffic Engineering-II: Traffic Studies,
Parking study, Accident Studies, Statistical Analysis of Traffic Data, Road
Safety Audit 2 Traffic Engineering-III: Highway Capacity and Level of Service 2
Traffic Engineering-IV: Intersection Control and Signalization 6 Railway
Engineering: Introduction, Geometric design of railway track - Gradients,
Circular Curves, Cant, Cant Deficiency, Transition Curves, Vertical Curves,
Widening of Gauge on Curves; discussion of salient features of high speed rail
tracks - 138 - 2 Airport Engineering: Introduction to Air transportation,
airport layout and components; 4 Runway and taxiways: Runway - orientation,
wind rose diagrams, geometric standards, basic runway length, correction of
runway length; taxiway: geometric standards and high speed exit taxiway design.